

KHALIL OUALDI

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Education

EPFL — BSc in Mathematics, Section de Mathématiques

Sept 2025 – June 2028 (expected)

- Cumulative average **5.13/6**; propaedeutic (first) year **5.63/6**, passed at first attempt.
- Highest grades: Advanced Linear Algebra I (**6.0/6**), Information & Computation (**6.0/6**, C++), Advanced Analysis I (**5.75/6**), Object-Oriented Programming (**5.25/6**, Python).
- Completed: Advanced Analysis I–II, Advanced Linear Algebra I–II (Jordan normal form, SVD, Eckart–Young low-rank approximation), Algebra I, Discrete Mathematics & Combinatorics, two programming courses.
- Autumn 2026 (3rd semester): Probability Theory (Prof. M. Hairer), Analysis III (Fourier analysis, Lebesgue measure theory), Topology, Differential Geometry, Algebra II.

Lycée Pierre Mendès France, Tunis — Baccalauréat Français International (British section)

2019 – 2025

- Ranked **1st of 300**; overall average **19.36/20**; *Félicitations du jury*.
- Majors: Mathematics, Physics–Chemistry, Engineering Science; option *Mathématiques Expertes*.

Teaching & Leadership

Teaching Assistant, MATH-110(a) Advanced Linear Algebra I — EPFL

Sept 2026 – present

- Selected for a paid assistantship in Prof. Daniel Kressner's teaching team on the strength of a 6.0/6 in the course; 6 h/week (3 h of exercise sessions, 3 h of preparation and grading).
- Lead two weekly exercise sessions for ~40 students each on vector spaces, multilinear forms and determinants; grade submitted problem sets and write up model solutions.

Coach, Coaching Association of the Section of Mathematics — EPFL

2026 – present

- Organise leisure activities and integration events for incoming students in the Section of Mathematics, helping first-years settle into the section socially over their first year.

Founder and Lead Instructor, OlymPMF competitive mathematics club — Tunis

2022 – 2025

- Founded and ran a weekly 4-hour session (2 h lecture, 2 h tailored problem set) in olympiad mathematics and algorithmics; grew the club from **10 to over 50 members** across all high-school years.
- Wrote every piece of teaching material from scratch — 2–3 pages of lecture notes plus a matched exercise sheet per session — covering graph theory, combinatorics, algebra, analysis and geometry.
- Coached **5 members to podium finishes** for Tunisia at AEFÉ-organised mathematics competitions.
- Handed the club and its full course archive to two successor students on graduation; membership has since passed 70 and the initiative became the school's permanent *labo-maths*.
- Asked by the school council and the mathematics faculty to provide targeted exam-preparation support outside the club.

Projects

Adventure — 2D action game in Python, team of two

2026

- Built a navigation-mesh path-finding system (A* with octile heuristic) over a 3×3 sub-node grid with integer-coded nodes, driving autonomous enemy pursuit of the player.
- Implemented finite-state-machine enemy AI, an abstract weapon hierarchy (sword, boomerang) and animation logic; Python with arcade and uv.
- Coordinated development with a partner on GitHub; largest codebase written to date.

Les Petits Trains — simulation game in C++

2026

- Designed and implemented the full simulation in C++ using standard containers and algorithms, as the term project of the Information & Computation course.

Carbon footprint of the Formula 1 ecosystem — research paper, EPFL

2026

- Co-authored a quantitative study of emission-reduction levers across the F1 logistics chain; written in $\text{\LaTeX}$ and coordinated with a partner over Git.

Technical Skills

Programming: C++ (data structures, algorithms, competitive problem solving), Python (object-oriented design, simulation).

Scientific computing: NumPy, Matplotlib; currently working through a structured PyTorch deep-learning curriculum.

Tools: Git/GitHub (used throughout coursework and team projects), $\text{\LaTeX}$ (proof write-ups and reports), Linux.

Mathematics: linear algebra and matrix decompositions (SVD, spectral theory), analysis, probability, combinatorics and graph theory.

Honours & Mathematical Competitions

- **Eight gold medals**, Tunisian national mathematical olympiads (ATSM / ATTCM), 2022–2024; six consecutive years in the national olympiad training programme, 2018–2024.
- **8th of 12** in the final selection round for the Tunisian team to the International Mathematical Olympiad of 2024.
- **2nd prize** (3rd score in the combined 5^e/4^e category), *Coupe de Printemps* — qualifying for the Animaths Valbonne camp, a two-week intensive olympiad training selecting **80 candidates from roughly 3,000**, 2022.
- Admitted through the *Coupe d'Automne* to **POFM**, the year-long French national olympiad preparation and team-selection programme (bimonthly courses, monthly selection tests), 2022–2023.
- **4th worldwide** across the AEFÉ network of French schools, Concours René Merckhoffer, 2021.
- Qualified with the Tunisian team for the finals of the 36th *Championnat International des Jeux Mathématiques et Logiques*, held at EPFL.

Languages

French — bilingual · Arabic — native · English — C1/C2 (certified, Baccalauréat Français International, British section) · Spanish — intermediate.